AMATH 445 Scientific Machine Learning

Reconstruction of excitation waves from mechanical deformation using physics-informed neural networks

Nathan Dermul and Hans Dierckx

Presented by Robin Zhang, Yassa Taiseer, and Jie Zhang

Background & Introduction

Cardiac arrhythmias are abnormal heart rhythms that affect millions of people worldwide and are responsible for a large proportion of sudden cardiac deaths. Accurate diagnosis and treatment of arrhythmias require identifying the electrical activation patterns that propagate through cardiac tissue. Traditionally, these activation patterns are measured using invasive catheter mapping, where electrodes are inserted into the heart. Although effective, this method is expensive, time-consuming, and poses potential risks to patients.

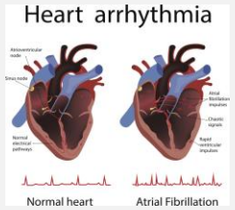
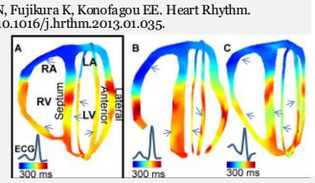
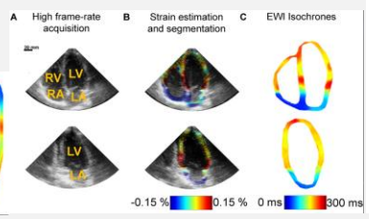


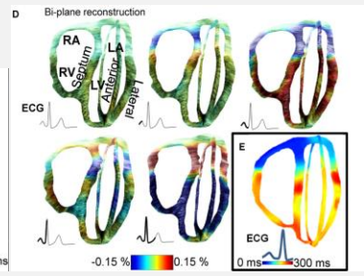
Figure: https://medicalterminologyblog.com/arrhythmia-dysrhythmia/

Recent advances in medical imaging have introduced electromechanical wave imaging (EWI) using ultrasound to measure the mechanical deformation of heart tissue during contraction. Since electrical excitation triggers mechanical contraction, it may be possible to infer electrical activity indirectly from these mechanical signals. However, reconstructing electrical activation from deformation data is a complex inverse problem because the mechanical response is influenced by spatial coupling and tissue properties. In this study, we investigate the use of Physics-Informed Neural Networks (PINNs) to reconstruct excitation wavesfrom mechanical deformation data. PINNs integrate machine learning with physical laws by incorporating governing equations directly into the neuralnetwork training process. By combining mechanical modeling and neural networks, this method provides a promising approach for non-invasive cardiac electrophysiology mapping, which could improve arrhythmia diagnosis, reduce clinical risk, and enable better treatment planning.

Figures: Provost J, Lee WN, Fujikura K, Konofagou EE. Heart Rhythm. 2013;10(6):856–863. doi:10.1016/j.hrthm.2013.01.035.







Methods

Generation of Synthetic Data

Forward Model

The synthetic data set was generated by modeling electrical activation waves that propagate through a 2D elastic medium. A modified Aliev-Panfilov kinetics are used to model the excitation wave dynamics.

$$\frac{\partial v}{\partial t} = \Delta v - kv(v-a)(v-1) - vw$$

$$\frac{\partial w}{\partial t} = \epsilon(v)(kv - w)$$

Next, the transmembrane voltage v induces active tension T_A in the medium according to:

$$\frac{\partial T_a}{\partial t} = \epsilon(v)(k_T v - T_a)$$

Lastly, Navier–Cauchy equations for mechanical equilibrium in an elastic medium are used.

$$(\lambda + \mu)\,\vec{\nabla}(\vec{\nabla}\cdot\vec{U}) + \mu\Delta\vec{U} = -\vec{\nabla}T_a$$

Simulations

Simulated focal and spiral cardiac waves, converting voltage v into active tension T_a and deformation (U_x, U_y) yielding synthetic 40×40 data over 30 time steps.

Reduction of Data Quality

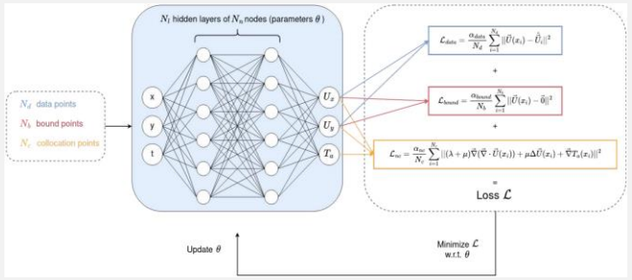
1) Gaussian noise
2) Lower spatial resolution by averaging over $n \times n$
3) Tri-planar sparse sampling (mimicking ECG)

Quality Assessment of the Reconstruction

Use FSIM (Feature Similarity Index).

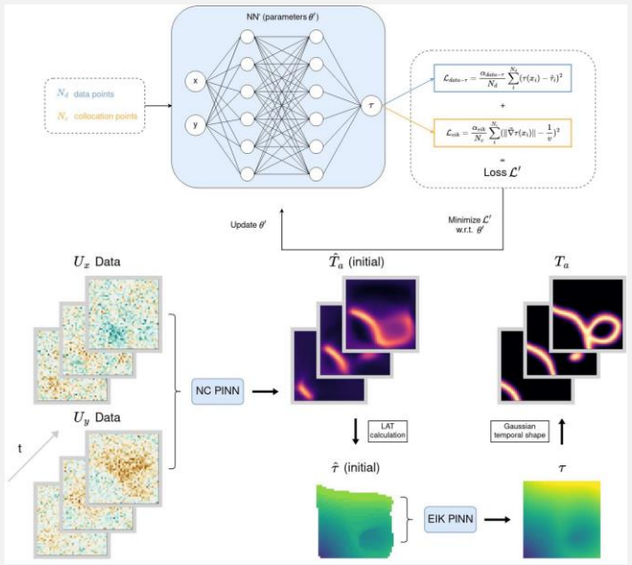
Inversion Model

Navier-Cauchy Physics-Informed Neural Network (NC-PINN)



Additional Regularization

Use a second PINN based on the eikonal equation to refine the activation-time map after the initial NC-PINN reconstruction, which is then converted back into a cleaner active-tension wave.



Results

Inversion using the NC-PINN

Under ideal conditions with full spatial-temporal data and no noise, the NC-PINN works well for reconstructing both focal and spiral active-tension patterns from deformation data.

- Reasonable run time of ~200s.
- Largest error near sharp gradients.
- Reconstructed wave is too symmetric
- Border amplitudes are reduced by the BC
- Spiral becomes blurrier later on.

Effect of noise

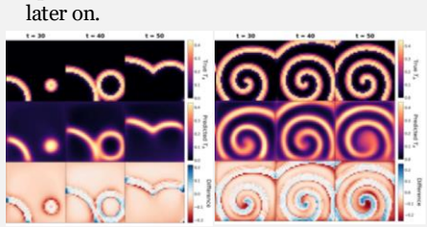
Effect of noise worsen the reconstruction quality, and 1) the focal pattern remains recognizable even at 50% and 2) spiral pattern degrades much faster beyond 30%.

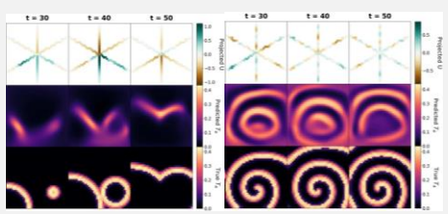
Effect of spatial resolution

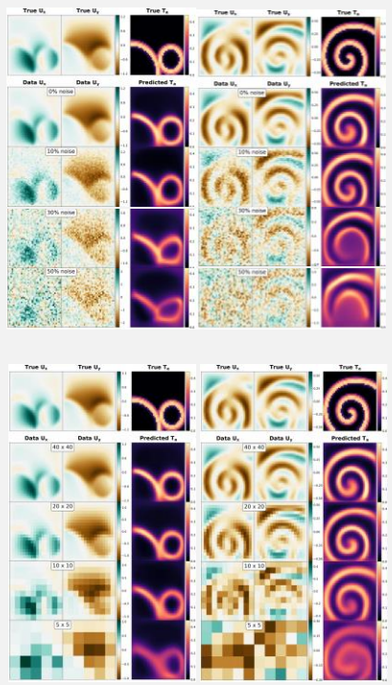
Reconstruction is quite robust under moderate spatial down sampling (10 s.u.), but when the data become very course (20 s.u., the main pattern is still visible but sharp boundaries are lost, and the spiral deteriorates more than the focal one.

Effect of tri-planar projections

The model can still recover some global activation structure even from this very sparse input, but find details are lost, border/early-time features are missed, and although spiral struct seems to still exist, it is becoming difficult to confidently distinguish a spiral from a focal source.

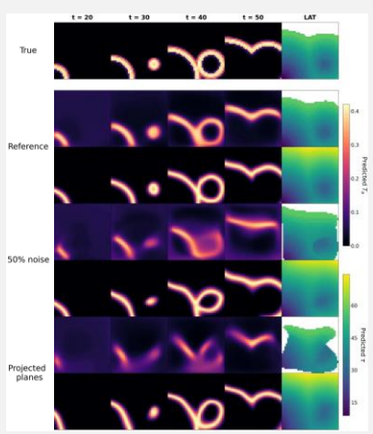






Inversion using wave propagation

The NC+EIK method improves reconstruction mainly when the data is very poor, especially in the sparse tri-planar case. When data is clean, it performs similarly to the basic NC-PINN because the original reconstruction is already accurate.



Discussion & Conclusion

Overall, this work shows that Physics-Informed Neural Networks (PINNs) can bridge the gap between electrical causes in cardiac tissue and mechanical observations. Even with sparse or noisy data, the framework remains reasonably robust to reconstructs both focal sources and intricate spiral waves.

By enabling physicians to create high-fidelity electrical maps from standard ultrasound, this research may open the door to a non-invasive clinical pipeline, lowering patient risk and enabling more accurate, individualized cardiac disease treatment plans by enabling physicians to replace invasive catheter mapping.

References

Dermul, N., Dierckx, H. Reconstruction of excitation waves from mechanical deformation using physics-informed neural networks. Sci Rep 14, 16975 (2024). https://doi.org/10.1038/s41598-024-67597-3